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A fault diagnosis method for planetary gear under multi-operating conditions based on adaptive extended bag-of-words model

Hao Zheng [a], Gang Cheng [a,b,*], Yong Li [a], Chang Liu [a]

[a] School of Mechatronic Engineering, China University of Mining and Technology, 221116 Xuzhou, China
[b] Jiangsu Engineering Research Center of Intelligent Mechanized Mining Equipment, China University of Mining & Technology, China

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ABSTRACT

The traditional bag-of-words (BoW) model needs to set the number of basic words manually, and is not suitable for the fault diagnosis under multi-operating conditions. To solve these problems, a fault diagnosis method for planetary gear under multi-operating conditions based on adaptive extended bag-of-words (AEBoW) model is proposed. Defining the sum of squared errors (SSE) which describes intra-class dispersion, the optimal number of basic words is obtained by the gradient descent of SSE. The BoW model is extended to three layers, and the codebooks in each layer are constructed by extracting local kurtosis (LK), local 2-dimensional information entropy (L2DIE) and Unoriented-SIFT features respectively. The final diagnostic result is obtained by screening the classification results with higher probability layer-by-layer. The overall recognition rate is 99.5% for 25 fault states (5 rotating speeds and 5 gear fault types), and the accurate recognition rate of gear fault type is 100%.

1. Introduction

Planetary gear is widely used in aerospace, automobile, wind power generation and heavy industry due to its advantages of large transmission ratio, strong load-carrying capacity and high transmission efficiency [1]. However, planetary gear usually works in harsh environments, and its key components are prone to failure [2]. These faults lead to security incidents, so it is of great significance to carry out the research on fault diagnosis of planetary gearbox [3]. For a long time, the research of fault diagnosis methods for planetary gear has mainly focused on the fault diagnosis under single operating condition. In actual work, the operating conditions of the planetary gear are usually in a state of change, such as changes in the rotating speed and load [4]. The characteristic distribution of vibration signals is different at different rotating speeds [5], so it is difficult to diagnose the faults of planetary gear at different rotating speeds through one model [6]. Therefore, it is of great practical significance to study a model that is suitable for the fault diagnosis for planetary gear at different rotating speeds.

The vibration signals of planetary gears have strong characteristics of non-linearity, non-stationarity and weak fault features [7–9]. To solve these problems, some scholars have proposed time-frequency analysis methods to describe non-stationary signals by extracting frequency features at different times, such as Wigner-Ville decomposition (WVD) [10], wavelet transformation (WT) [11], local mean decomposition (LMD) [12], empirical mode decomposition (EMD) [13]. However, the performance of traditional time-frequency methods is not stable for nonlinear signals with weak fault features. Therefore, many scholars have proposed improved feature extraction methods for planetary gear vibration signals. Liu et al. [14] proposed an innovative soft sifting stopping criterion for improving LMD, and the improved LMD was used in fault diagnosis of planetary gearboxes. Chen et al. [15] used ensemble empirical mode decomposition (EEMD) to decompose planetary gearbox vibration signals, and then Teager-Kaiser energy operator was used to track the instantaneous amplitude and frequency of a mono-component signal. Zhao et al. [16] combined dynamically weighted wavelet coefficients and deep residual networks to improve diagnostic performance for planetary gearboxes. Li et al. [17] used adaptive multi-scale morphological filter (AMMF) to remove the fault-unrelated components, and used modified hierarchical permutation entropy (MHPE) to extract the fault features from the de-noised vibration signals of planetary gearboxes.

It is obvious that feature extraction methods for planetary gears are becoming more and more complex, and the reason for which is that those methods highlight fault features by de-noising. Researchers have devoted great effort to noise reduction [18–20]. However, in front of weak fault features and complex noises of

* Corresponding author at: School of Mechatronic Engineering, China University of Mining and Technology, 221116 Xuzhou, China.

E-mail address: zhenghao065219@163.com (G. Cheng).

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planetary gear vibration signals, it is obviously difficult to highlight fault features by de-noising. Therefore, Zheng et al. [21] proposed FAST-Unoriented-SIFT feature extraction method. This method extracts a large number of features directly from original signals without de-noising process. In this paper, based on FAST-Unoriented-SIFT, local kurtosis (LK) and local 2-dimensional information entropy (L2DIE) are proposed. LK reflects impact features of vibration signals [22], while L2DIE reflects the complexity of vibration signals [23]. The planetary gear vibration signals are well described by calculating different LK and L2DIE features. These feature extraction methods without noise reduction can greatly improve the efficiency of feature extraction.

In recent years, dictionary model has been widely used in representing vibration signals [24–26]. A redundant dictionary is built based on a kind of features, and then the signal is represented by linearly fitting of each word in the dictionary [27]. However, the linear fitting is time-consuming and has limited ability of representation. The bag-of-words (BoW) model represents signals by counting the frequency of each word, which has higher operating efficiency and stronger ability of representation [28]. Therefore, BoW model is widely used in text categorization and image recognition [29–31]. Recently, some scholars have applied BoW model to fault diagnosis. Jia et al. [32] extracted features from infrared thermography of bearing to build a BoW model, and the superiority of BoW model in fault diagnosis was illustrated by comparing with convolution neural network. Zheng et al. [21] converted the vibration signals into gray images, and then extracted features from images to establish a BoW model.

However, the number of basic words in the BoW model needs to be set manually, which is time-consuming and labor-consuming, and makes the portability poor. In addition, with the increase of fault types, the number of codebook vectors in BoW model is also increasing, which reduces the efficiency and accuracy of recognition. Therefore, BoW model is not suitable for fault diagnosis under multi-operating conditions.

In order to solve the above problems, a fault diagnosis method for planetary gear under multi-operating conditions based on adaptive extended bag-of-words model (AEBoW) is proposed in this paper. The sum of squared errors (SSE) is defined to describe the intra-class dispersion, and the optimal number of basic words is obtained by the gradient descent of SSE. In addition, in order to diagnose faults under multi-operating conditions through one model, the BoW model is extended to three-layer structure, using LK, L2DIE and Unoriented-SIFT features to build codebooks for each layer. In each layer, the probabilities that a signal belongs to each fault type are calculated, and the fault types with higher probability are transferred to the next layer, which are the prospective classification results in the next layer. Finally, the diagnostic result is obtained by layer-by-layer screening.

The main contributions of this paper are concluded as follows:

- AEBoW model is proposed, which realizes the adaptive selection of the number of basic words, and is suitable for fault diagnosis of planetary gear under multi-operating conditions.
- LK and L2DIE features are proposed, which extract a large number of features quickly without de-noising process.
- The overall recognition rate is 99.5% for 25 fault states (5 rotating speeds and 5 gear fault types), and the accurate recognition rate of gear fault type is 100%.
- The proposed method has high operating efficiency, which meets the real-time requirement, and has good application prospects.

The remainder of this paper is as follows: In Section 2, the mathematical model of the fault diagnosis method based on AEBoW model is established. In Section 3, the fault simulation experiment of planetary gear under multi-operating conditions is performed, and the data is collected. In Section 4, the validity of the proposed method are verified. In Section 5, the advantages of AEBoW model in recognition rate and operating efficiency are proved through comparative experiments, and the selection of gray level is discussed. In Section 6, the conclusion is stated.

2. Model building

2.1. Feature extraction

Firstly, the raw vibration signals are transformed into gray images, and then feature points are detected. Finally, the LK, L2DIE and Unoriented-SIFT features of each feature point are calculated respectively. The specific algorithms are as follows:

2.1.1. Signal conversion and feature points detection

Firstly, the vibration signals are transformed to 256-level gray images, which means the gray-scale value of each pixel is in the range of [0,255]. Therefore, the amplitudes of the vibration signal $a = \{a_1, a_2, \cdots, a_n\}$ (n is the signal length) are normalized to this range according to formula (1):

$$a_{norm} = \frac{127}{\max(|a|)} \cdot a + 128 \tag{1}$$

where a_{norm} is the normalized vibration signal, $\max(|a|)$ is the maximum absolute value of a. The amplitudes of vibration signals are normalized to [1,255] by the calculation of Eq. (1). Then, a_{norm} is reshaped into a matrix A with h rows and l cols ($l \times h = n$) according to formula (2):

$$A_{h'l'} = a_{norm}(l \cdot (h' - 1) + l'), \quad h' \in \{1, 2, \ldots, h\}, l' \in \{1, 2, \ldots, l\} \tag{2}$$

where $A_{h'l'}$ presents the element at row h' and column l' of A, and $a_{norm}(i)$ is the i-th value of a_{norm}. In order to make the texture features of the image more obvious, l and h should be similar. Ref. [21] has shown that the texture features of gray images with different fault types are different. Finally, the gray image is obtained by taking each element value of A as the gray-scale value of each pixel, as shown in Fig. 1 (Vibration signal to Gray image).

Afterwards, the FAST algorithm proposed by Edward Rosten [33] is used to detect the feature points. As shown in Fig. 1 (Feature points), the color circles represent the detected feature points. A feature point is used to illustrate FAST algorithm. Take the pixel p as the center, take a circle with radius 4, and there are 16 pixels on the circle $x \in \{1, 2, \ldots, 16\}$. Calculate the state of each pixel x relative to the p as follows:

$$S_{p\to x} = \begin{cases} d, & I_{p\to x} \leqslant I_p - t \quad \text{(darker)} \\ s, & I_p - t \leqslant I_{p\to x} \leqslant I_p + t \quad \text{(similar)} \\ b, & I_p + t \leqslant I_{p\to x} \quad \text{(brighter)} \end{cases} \tag{3}$$

where $I_{p\to x}$ is the gray-scale value of pixel x around p, and I_p is the gray-scale value of p. d indicates x is darker than p, s indicates x is similar to p, and b indicates x is brighter than p. According to Rosten's definition, if the states of more than 12 pixels in $x \in \{1, 2, \ldots, 16\}$ are d or b, p is considered as a feature point. According to the recommendation of Ref. [21], the threshold t = 30.

2.1.2. LK feature

Centrifugal force formula:

$$f = mv^2/r \tag{4}$$

where m is the mass of the particle, v is the rotating velocity, r is the radius of rotation and f is the centrifugal force on the particle.

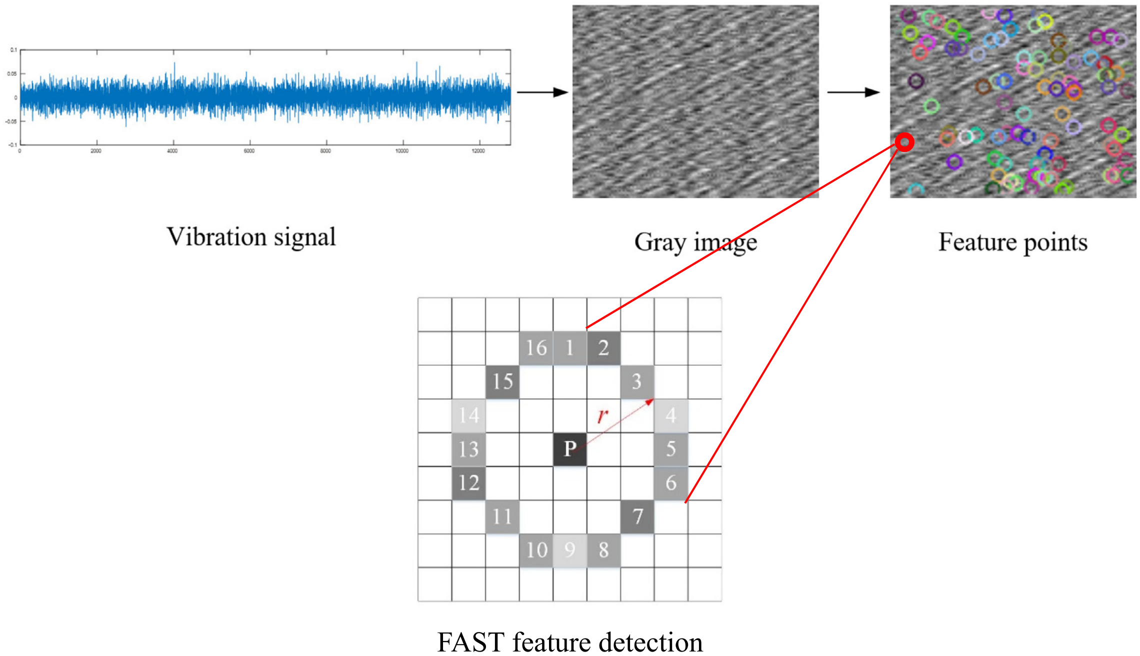

Fig. 1. Signal conversion and feature points detection.

According to Eq. (4), the centrifugal force on a particle is proportional to the square of the rotating speed, so the vibration amplitude measured by the acceleration sensor is also related to the square of the rotating speed. Therefore, the average amplitude of a signal is calculated as an amplitude factor:

$$\bar{a} = \frac{1}{n}\sum_{i=1}^{n}|a_i|, i = 1, 2, \ldots, n \tag{5}$$

where n is the number of sampling points and a_i is the vibration amplitude at point i.

Taking the square area with the feature point p as the center and w as the side length as a window, the LK feature in the window is defined as:

$$k_p = \bar{a} \cdot \frac{1}{w^2}\sum_{i=1}^{w}\sum_{j=1}^{w}\frac{(I_{ij}-\mu)^4}{\sigma^4} \tag{6}$$

where k_p is the kurtosis of point p, and $\bar{a}$ is the amplitude factor calculated in Eq. (5). I_{ij} is gray-scale value of the pixel at row i and column j in the window. μ is the mean gray-scale value in the window $\mu = \frac{1}{w^2}\sum_{i=1}^{w}\sum_{j=1}^{w}I_{ij}$, σ is the standard deviation $\sigma = \sqrt{\frac{1}{w^2}\sum_{i=1}^{w}\sum_{j=1}^{w}(I_{ij}-\mu)^2}$. Because the window size of Unoriented-SIFT feature (Section 2.1.4) is 16×16, in order to calculate three features simultaneously in the same window, w = 16 in Eq. (6).

2.1.3. L2DIE feature

The L2DIE feature in the same window is defined as:

$$e_p = \bar{a} \cdot \sum_{i=1}^{w}\sum_{j=1}^{w}(-F_{ij}\log F_{ij}) \tag{7}$$

where e_p is the L2DIE of point p, and $\bar{a}$ is the amplitude factor calculated in Eq. (5), and F_{ij} is the frequency of the gray-scale value at row i and column j appearing in the window. For the same reason as Eq. (6), w = 16 in Eq. (7).

2.1.4. Unoriented-SIFT feature

According to the Unoriented-SIFT algorithm proposed in Ref. [21], the 16×16 window centered on feature point p is divided into 16 4×4 squares. Then, according to Eqs. (8) and (9), the modulus and direction of the gradient of each pixel are calculated respectively:

$$m(i,j) = \sqrt{[I(i+1,j) - I(i-1,j)]^2 + [I(i,j+1) - I(i,j-1)]^2} \tag{8}$$

$$\theta(i,j) = \arctan\frac{I(i,j+1) - I(i,j-1)}{I(i+1,j) - I(i-1,j)} \tag{9}$$

where $I(i,j)$ is the gray-scale value of the pixel at row i and column j in the window, $m(i,j)$ is the modulus of the gradient, and $\theta(i,j)$ is the direction of the gradient. The cumulative gradients of 8 directions (0–360 degrees at intervals of 45 degrees) are counted in each 4×4 square, then 8-dimensional vectors of each square are obtained. Finally, a $16 \times 8 = 128$ dimensional vector is obtained on 16 squares as the Unoriented-SIFT descriptor of feature point p.

2.2. AEBoW model

There are three main steps to establish the traditional BoW model:

Step 1: Feature extraction.
Step 2: Getting basic words. Usually, K-Means [34] clustering algorithm is used to aggregate the extracted features into k clusters, and k clustering centers are used as the basic words.
Step 3: Constructing codebook. The frequency of each basic word in each fault type signal is counted to form a word frequency vector. Each word frequency vector is a description of a fault type, and all word frequency vectors constitute the codebook.

2.2.1. Adaptive selection of the number of basic words

In the Step 2 of building the traditional BoW model, K-Means takes k as a parameter and aggregate all features into k clusters $\{C_1, C_2, \ldots, C_k\}$, where k needs to be set manually. If k is too large, it means the amount of basic words is too large, which affects the efficiency of classification. If k is too small, the number of basic words is not enough to distinguish different fault types. Therefore, define SSE (sum of squared errors) describing intra-class dispersion as follows:

$$SSE = \sum_{i=1}^{k} \sum_{x \in C_i} |x - m_i|^2 \tag{10}$$

where x is the sample in cluster C_i, m_i is the cluster center of C_i (mean of all samples in cluster C_i).

SSE describes the degree of intra-class dispersion, so smaller *SSE* means better the clustering effect. In addition, *SSE* decreases with the increase of k, but the decline rate decreases gradually. When the descending curve of *SSE* tends to be flat, the optimal number of basic words k is obtained. In this paper, when *SSE* decreases less than 0.001 at $k = k_0$, and within the next 10 steps, *SSE* decreases no more than 0.01 at each step, then k_0 is the optimal number of basic words.

2.2.2. The structure and diagnosis method of AEBoW model

In order to make the BoW model suitable for the fault diagnosis of planetary gear under multi-operating conditions, the BoW model is extended to three-layer structure.

2.2.2.1. The structure of AEBoW model. First layer of the model:

(1) Extracting LK features. Firstly, the feature points are detected by the method in Section 2.1.1, and then the LK values of each feature point are calculated by the method in Section 2.1.2.
(2) Getting LK basic words. Because the LK feature is a numerical value, K-Means clustering is not necessary. Moreover, the distribution of LK values is relatively concentrated, so each LK value is taken as a basic word. The calculating accuracy of LK value is to retain two digits after decimal point.
(3) Constructing LK codebook. The frequency of each LK basic word in each fault type is counted, and the LK codebook is obtained.

Second layer of the model:

(1) Extracting L2DIE features. Firstly, the feature points are detected by the method in Section 2.1.1, and then the L2DIE values of each feature point are calculated by method in Section 2.1.3.
(2) Getting L2DIE basic words. For the same reason as LK basic words, each L2DIE value is taken as a basic word. The calculating accuracy of L2DIE value is to retain two digits after decimal point.
(3) Constructing L2DIE codebook. The frequency of each L2DIE basic word in each fault type is counted, and the L2DIE codebook is obtained.

Third layer of the model:

(1) Extracting Unoriented-SIFT features. Firstly, the feature points are detected by the method in Section 2.1.1, and then the Unoriented-SIFT feature of each feature point are calculated by the method in Section 2.1.4.
(2) Getting Unoriented-SIFT basic words. K-Means algorithm is used to aggregate all Unoriented-SIFT features into k clusters, and k clustering centers are the basic words. The optimal value of k is calculated by the method in Section 2.2.1.
(3) Constructing Uoriented-SIFT codebook. The frequency of each Unoriented-SIFT basic word in each fault type is counted, and the Unoriented-SIFT codebook is obtained.

2.2.2.2. The fault diagnosis method of a signal to be diagnosed.

(1) Feature extraction. Firstly, the feature points are detected from the signal to be diagnosed by the method in Section 2.1.1, and then the LK, L2DIE and Unoriented-SIFT features of each feature point are calculated by the methods in Section 2.1.2, 2.1.3 and 2.1.4 respectively.
(2) Basic words approximation. The features obtained in the previous step are replaced by the nearest LK basic word, L2DIE basic word and Unoriented-SIFT basic word respectively.
(3) Counting word frequency vector. The LK, L2DIE and Unoriented-SIFT word frequency vectors of the signal are counted respectively.
(4) The fault classification of first layer. The distances between the LK word frequency vector of the signal and the vectors in the LK codebook are calculated. The closer the distance is, the more similar the fault type represented by the LK codebook vector is. The probability of the fault type of the signal is calculated as following:

$$P_i = 1 - \frac{d_i}{\sum d_i}, \quad i \in \{1, 2, \ldots, N\} \tag{11}$$

where N is the total number of fault types, P_i is the probability that the signal belongs to fault i, and d_i is the distance between the word frequency vector of the signal and the vector representing fault i in the LK codebook. For $i \in \{1, 2, \ldots, N\}$, if $P_i > 98\%$, the i is added to the classification result R_1 of the first layer, and R_1 is also the prospective classification results of the second layer. If R_1 contains only one fault type, then the final diagnosis result is taken, otherwise it goes to the next layer.

(5) The fault classification of second layer. Calculate the distances between the L2DIE word frequency vector of the signal and the vectors representing the fault types of R_1 in L2DIE codebook, and calculate the probabilities of fault types of R_1 according to Eq. (11). For all i in R_1, if $P_i > 98\%$, the i is added to the classification result R_2 of second layer, and R_2 is also the prospective classification results of the third layer. If R_2 contains only one fault type, the final diagnosis result is taken, otherwise it goes to the next layer.
(6) The fault classification of third layer. The distances between the Unoriented-SIFT word frequency vector of the signal and the vectors representing the fault types of R_2 in the Unoriented-SIFT codebook are calculated, and the fault type represented by the nearest vector is taken as the final diagnosis result.

After obtaining the word frequency vectors of the signal to be diagnosed, the flow chart of the fault diagnosis is shown in the Fig. 2. In the Fig. 2, LIE means L2DIE, and US means Unoriented-SIFT.

In the process of fault diagnosis, LK, L2DIE and Unoriented-SIFT features are firstly extracted from the signal to be diagnosed. Then, each features are replaced by basic words like the description in Section 2.2.2.2 step 2 to form the word frequency vectors. Afterwards, the distances between each type of word frequency vector and the vectors in each kind of code book are calculated, and one or more of the most probable fault types are the classification results R_1, R_2 and the final result of classification. The probabilities of fault types are calculated through Eq. (11). If there is only one

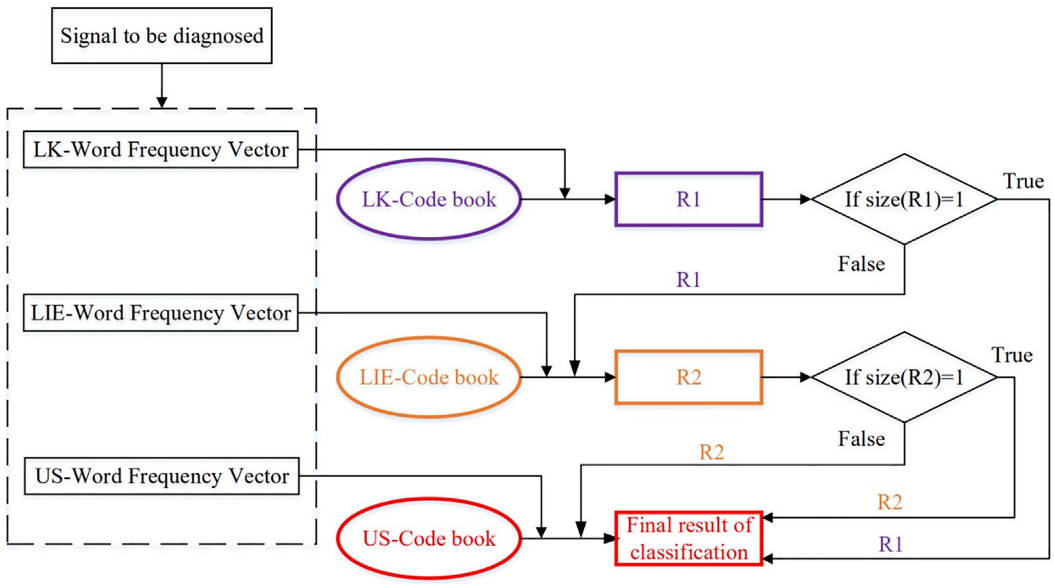

Fig. 2. Flow chart of the fault diagnosis of a signal to be diagnosed.

fault type in R_1 or R_2, the fault type is the final result of classification.

2.3. The fault diagnosis method based on AEBoW model

The fault diagnosis method based on AEBoW model consists of two steps:

Step 1: building AEBoW model based on training set as Section 2.2.2.1.
Step 2: fault diagnosis for test set as Section 2.2.2.2.

The fault diagnosis method based on AEBoW is shown in Fig. 3. In the Fig. 3, LIE and US mean L2DIE and Unoriented-SIFT respectively. The LK, LIE and US features of the test set need to be replaced by the nearest basic words of LK, LIE and US to obtain the word frequency vectors like the description in Section 2.2.2.2 step 2.

3. Fault simulation experiment

The simulation experiments of planetary gear faults under multi-operating conditions were carried out on the drivetrain dynamics simulator (DDS) of Spectra Quest Company in the United States. The experimental platform includes a control motor, a planetary gearbox, an acceleration sensor, a fixed-axis transmission box, a magnetic brake, a data acquisition instrument and a PC, as shown in Fig. 4. The planetary gearbox is composed of two-stage planetary gear transmission. The fixed-axis transmission box is supported by rolling bearings and has two-stage straight-tooth transmission. Vibration signals are acquired through the acceleration sensor and data acquisition instrument, and motor speed and brake are controlled through PC. Because the sun gear has high speed and meshes with other gears at the same time, it is easier to get faults. Therefore, the 5 faults of the secondary sun gear were simulated, including: broken gear, normal gear, gear with tooth root crack, gear with one missing tooth and wear gear as shown in Fig. 5. In the experiment, when output frequency of the motor is 25 Hz, 30 Hz, 35 Hz, 40 Hz and 45 Hz, fault simulation experiments are carried out for the 5 gear fault types, so 25 fault states in total are obtained as shown in Table 1. During the experiment, the brake load is set to 20 Nm, the sampling frequency is 12800 Hz, and each sample has 972,800 data point.

4. Experimental analysis

In order to verify the validity of the fault diagnosis method for planetary gear under multi-operating conditions based on AEBoW model, the vibration signals obtained from the fault simulation experiment are put into the fault diagnosis model shown in Fig. 3.

(A) Signal conversion and division of training set and test set

In the fault simulation experiment, the vibration signal of each fault states has 972,800 data points, which is evenly divided into 76 segments. Each segment is converted into a 100×128 gray image by the method in Section 2.1.1. Sixty segments of each fault type are selected as a training set and 16 segments as a test set. Finally, $25 \times 60 = 1500$ training samples and $25 \times 16 = 400$ test samples are obtained.

(B) Building AEBoW model based on training set

The AEBoW model is established based on the training set as Section 2.2.2.

Only the third layer of the model needs to get the basic words through K-Means algorithm. Therefore, in the third layer, the optimal number of basic words k is calculated by the method in Section 2.2.1, and the result is k = 101. The gradient descent curve of *SSE* is shown in the Fig. 6.

In addition, in the process of constructing the AEBoW model, there are two points to be explained:

The first point is the calculating accuracy of LK and L2DIE values. The calculating accuracy is two decimal places in this paper, which is mainly determined by the range of LK and L2DIE values and the number of features of each segment. In this experiment, the average feature quantities of a segment signal is 127. Taking LK feature as an example, the range of LK values calculated for all training samples is [0.0173, 0.2605]. Too high calculation accuracy makes signals of the same kind show different, while too low calculation accuracy lacks ability to distinguish different signals. Five signals of broken gear at five rotating speeds are taken as

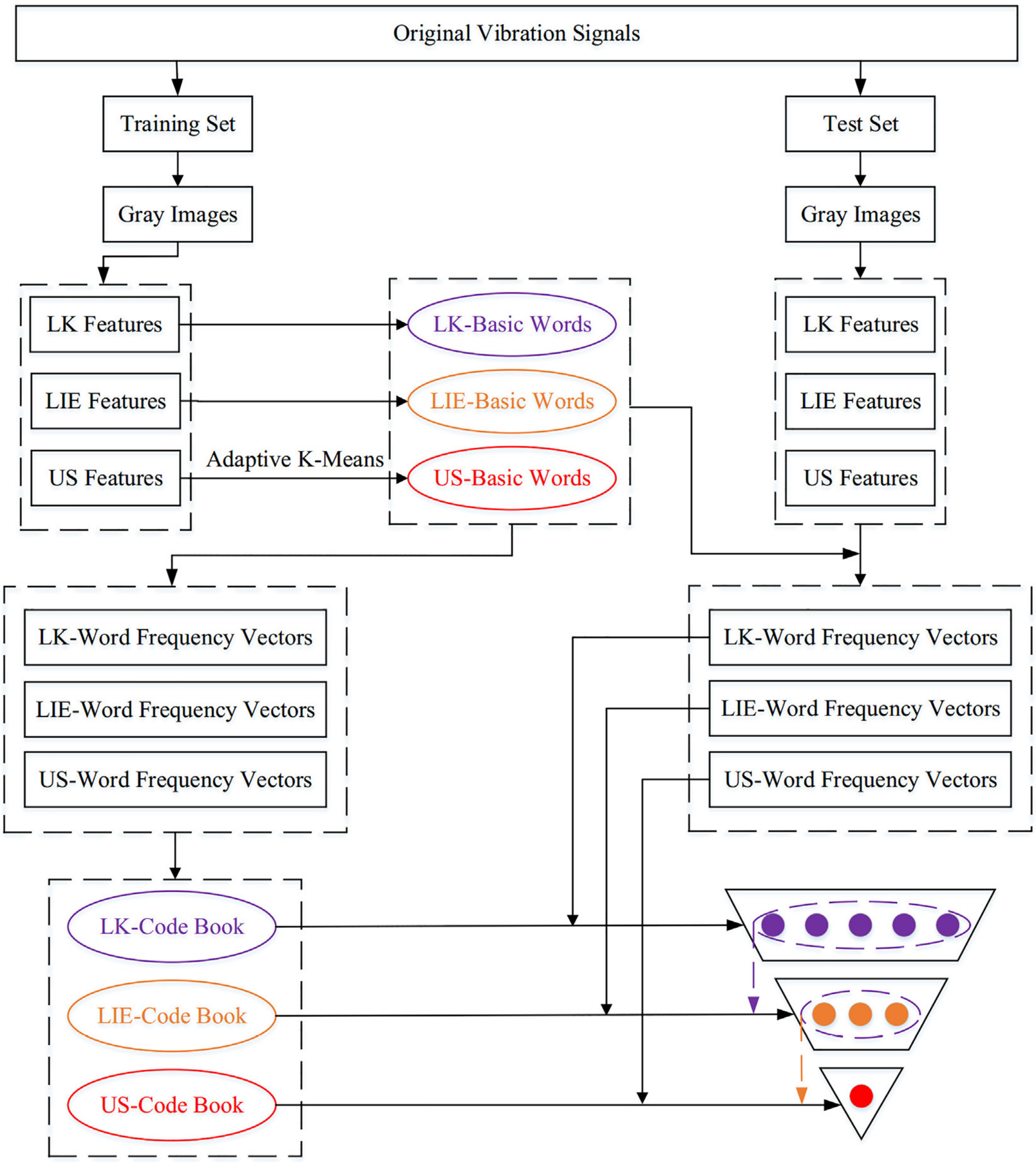

Fig. 3. Flow chart of the fault diagnosis method based on AEBoW model.

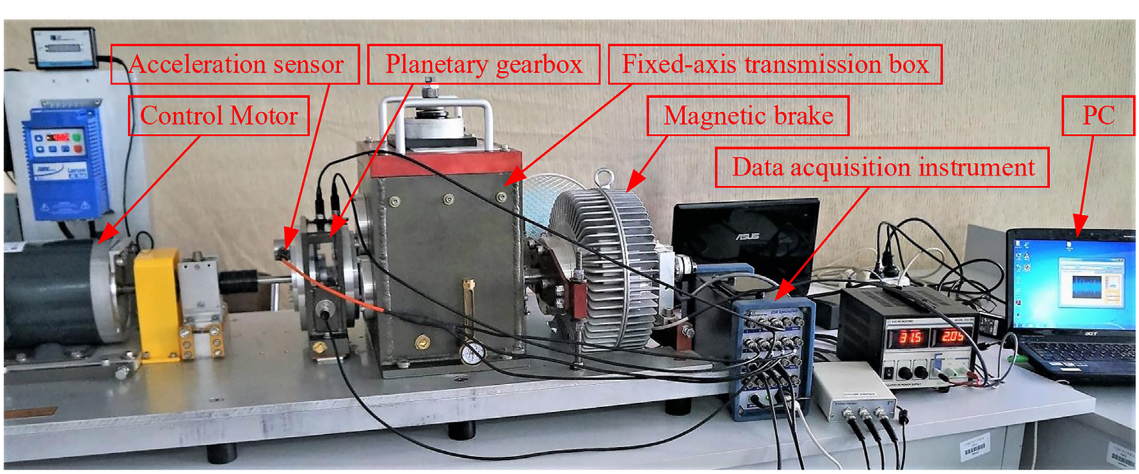

Fig. 4. The experimental platform.

Fig. 5. Five types of sun gear faults.

Table 1
25 fault states (5 rotating speeds and 5 gear fault types).

	25 Hz	30 Hz	35 Hz	40 Hz	45 Hz
Broken	S1-25	S1-30	S1-35	S1-40	S1-45
Normal	S2-25	S2-30	S2-35	S2-40	S2-45
Tooth root crack	S3-25	S3-30	S3-35	S3-40	S3-45
One missing tooth	S4-25	S4-30	S4-35	S4-40	S4-45
Wear	S5-25	S5-30	S5-35	S5-40	S5-45

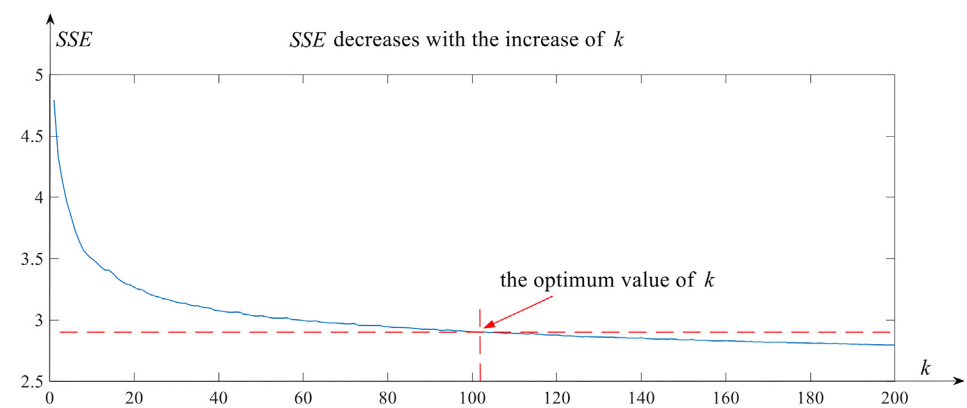

Fig. 6. The adaptive selection of the number of basic words.

experimental objects. When the calculation accuracy is one decimal place, only four basic words are obtained, ranging from 0 to 0.3. The word frequency vectors of the above five signals are: $[1, 0, 0, 0]$, $[0, 1, 0, 0]$, $[0, 1, 0, 0]$, $[0, 1, 0, 0]$, $[0.95, 0.05, 0, 0]$, which obviously lack ability to distinguish fault types. When the calculation accuracy is two decimal places, 28 basic words are obtained, ranging from 0 to 0.27. Five word frequency vectors of above five signals are shown in the Fig. 7.

Where the word frequency vectors show obvious differences. Therefore, it is not necessary to take three decimal places. For all training samples, the range of L2DIE is [0.0192, 0.1982]. For the same reason, the calculation accuracy of L2DIE is also two decimal places.

Another point need to be explained is why each layer only takes categories with probability greater than 98% as the classification result. The purpose of setting a threshold is to narrow down the

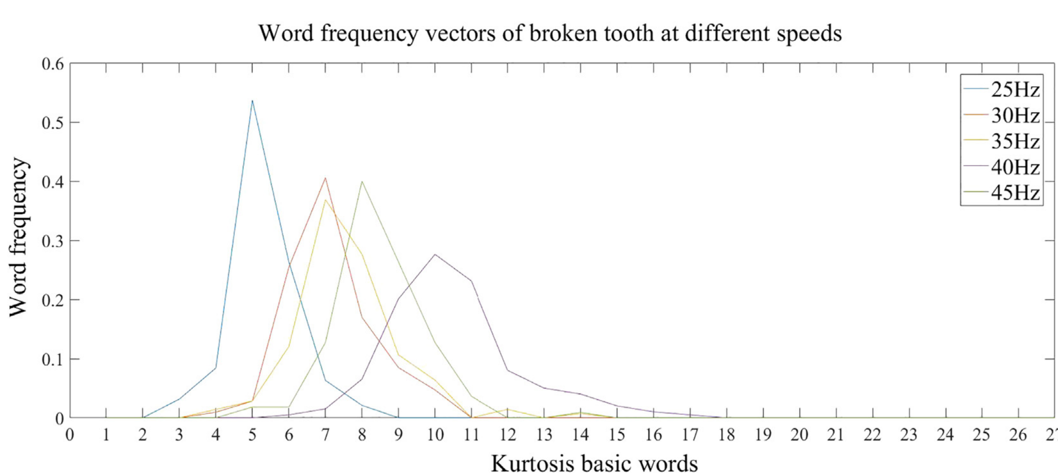

Fig. 7. 5 word frequency vectors of broken gear signals at 5 rotating speed.

codebook and prospective categories layer-by-layer, which improves the operating efficiency and diagnostic accuracy. In the first two layers of the model, it is necessary not only to ensure that prospective categories are as few as possible, but also to ensure that the correct category is included in the classification result. Therefore, the maximum threshold should be taken when the correct category is included. In the experiment, when the threshold is 99%, 67 of the 400 test samples in the first layer are not classified correctly. When the threshold is 98%, all the test samples are correctly classified in the first two layers. Therefore, 98% is selected.

(C) Fault diagnosis for test set

The fault diagnosis of each sample in the test set is carried out, and the results are shown in Fig. 8. Although classification results at different rotating speeds are drawn in different diagrams, actually samples at different rotating speeds are diagnosed together.

The abscissa of Fig. 8 represents 25 fault states of test samples. The numbers on the ordinate from 1 to 25 represent S1-25 to S5-45 respectively. The red dots and blue circles represent the actual categories and diagnostic categories of samples respectively. As shown in Fig. 8, 398 samples are correctly diagnosed, and the accuracy rate reaches 99.5%. The fault types of samples marked in red boxes are correctly recognized, but the rotating speeds of these samples are not. Therefore, without considering whether the rotating speed is accurately classified, the accurate recognition rate of gear fault types reaches 100%.

In 400 test samples, not every sample needs go through all three layers of the model. Some samples can be diagnosed in the first two layers. Fig. 9 shows the number of diagnosed samples in each layer.

As Fig. 9 shows, 18 and 27 samples are diagnosed in the first and second layer respectively. In total, 11.25% of samples are diagnosed in the first two layers, which means this model can reduce

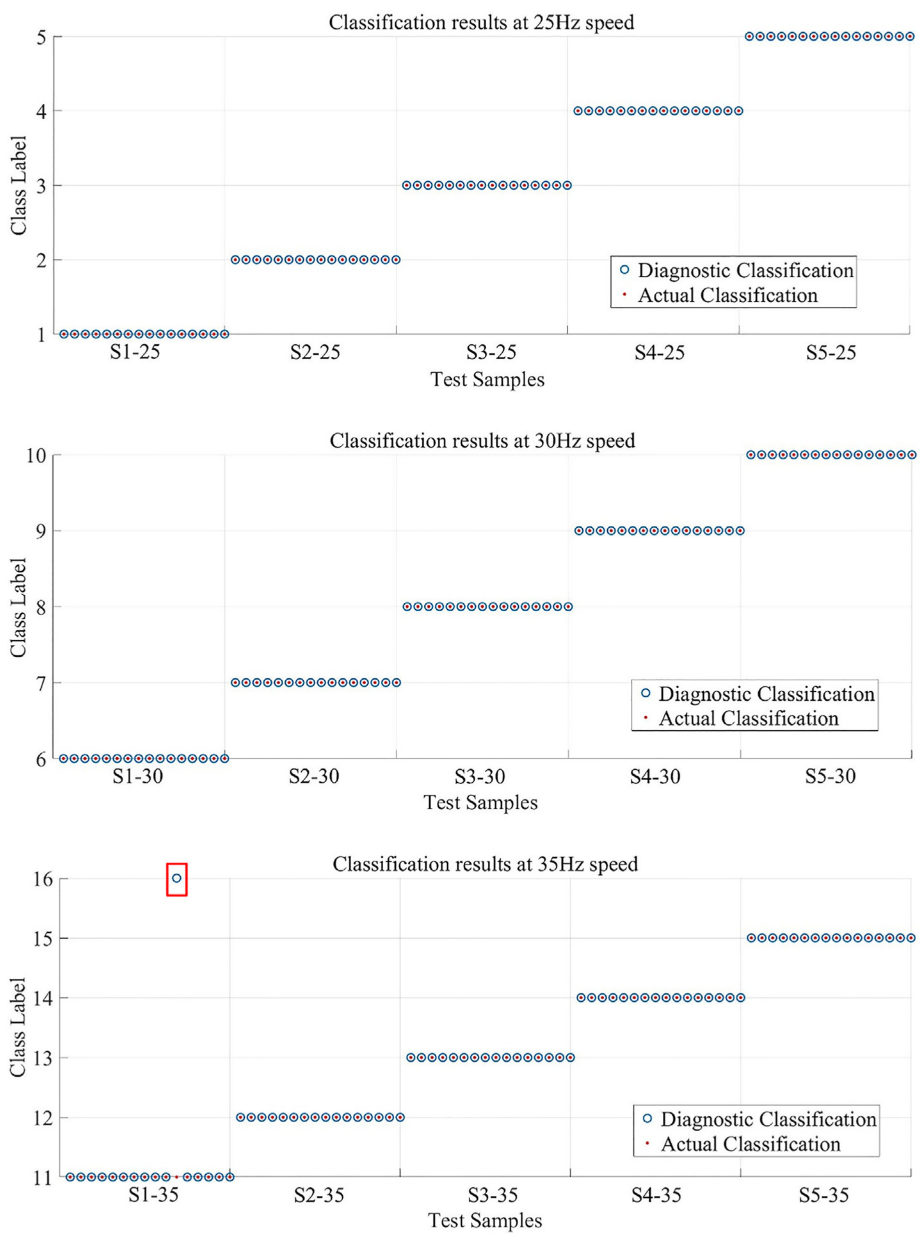

Fig. 8. Actual classification and diagnostic classification of test samples.

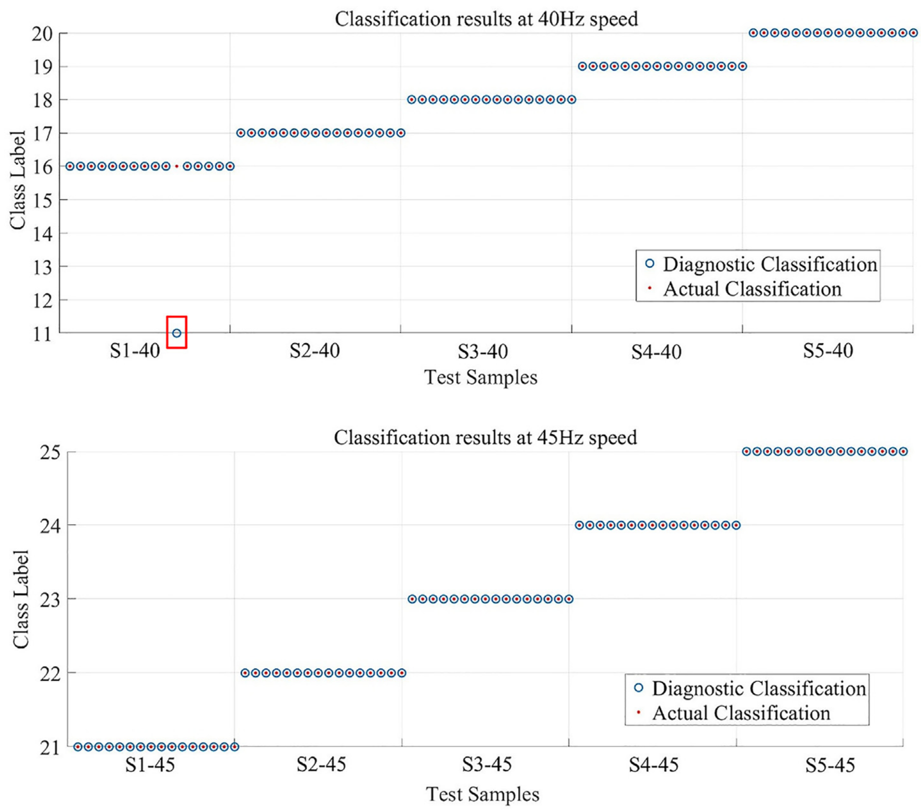

Fig. 8 (*continued*)

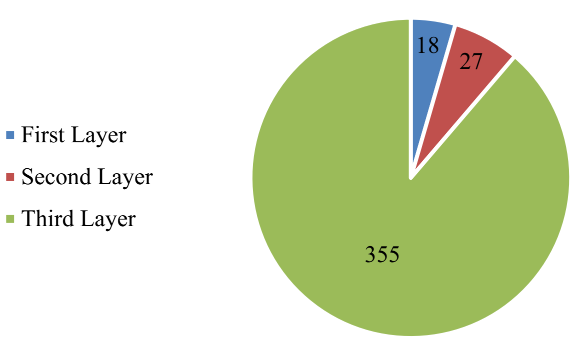

Fig. 9. The number of diagnosed samples in each layer.

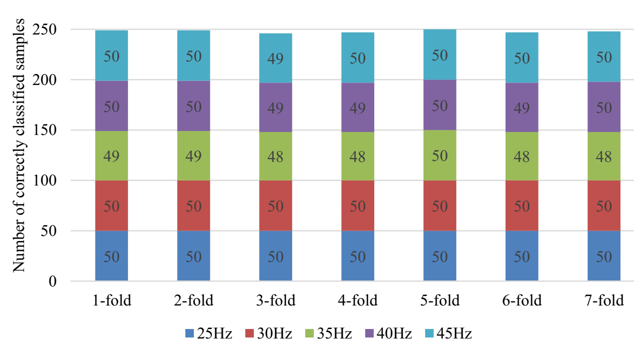

Fig. 10. Experimental results of 7-fold cross-validation.

the computation load and the possibility of error through layer-by-layer screening.

(D) 7-fold Cross-Validation

To further verify this method, a 7-fold cross-validation experiment was conducted. The first 70 samples in each fault state (25 states in total) were evenly divided into 7 groups. One group was taken as the test set each time, and the rest was taken as the training set. Fig. 10 shows the number of correctly classified samples in each experiment, and different colors represent different rotating speeds. There are 5 fault types at each rotating speed, and each type has 10 test samples.

As Fig. 10 shows, the recognition rates in each experiment are 99.60%, 99.60%, 98.40%, 98.80%, 100%, 98.80%, 99.20% respectively, and the average recognition rate is 99.2%. Therefore, the effectiveness of this method is validated.

Table 2
The number of diagnosed samples in each layer.

	1-fold	2-fold	3-fold	4-fold	5-fold	6-fold	7-fold	Average
1st layer	12	12	11	17	14	14	11	13
2nd layer	4	43	1	8	45	30	11	20.29
3rd layer	234	195	238	225	191	206	228	216.71

Furthermore, Table 2 shows the number of diagnosed samples in each layer. On average, 13.32% of samples are diagnosed in the first two layers.

5. Comparative analysis

5.1. Compared with BoW model

(A) Recognition rate

The diagnostic results of the traditional BoW model for the test samples are shown in the Fig. 11.

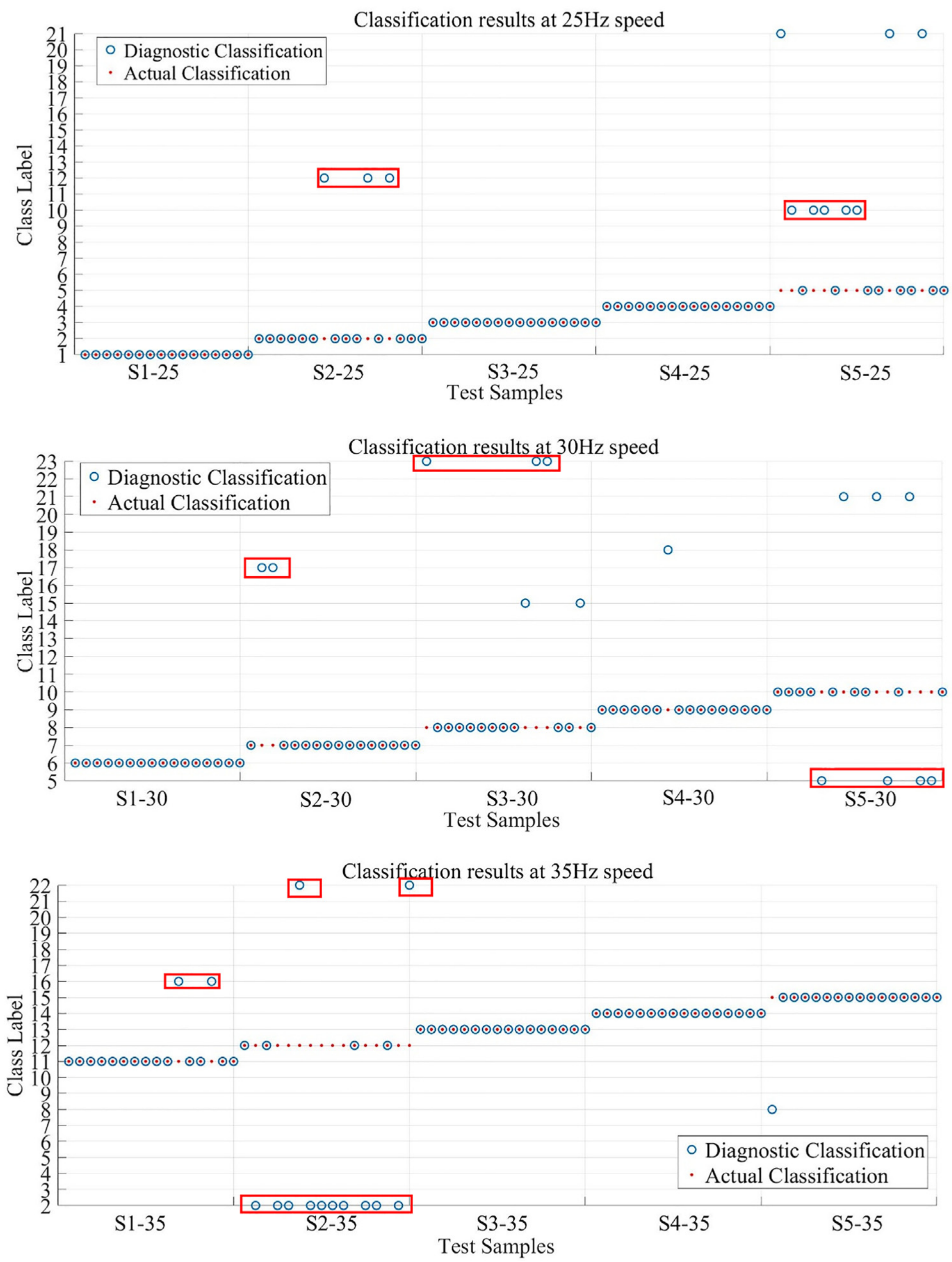

Fig. 11. Diagnostic results of BoW model for test samples.

The abscissa in Fig. 11 represents 25 fault states of test samples, and the numbers on the ordinate from 1 to 25 represent S1-25 to S5-45 respectively. The red dots and blue circles represent the actual categories and diagnostic categories of samples respectively. As shown in Fig. 11, 337 samples are correctly diagnosed, and the accuracy of fault diagnosis is 84.25%. The gear fault types of the samples marked in the red boxes are accurately recognized, but the rotating speeds of these samples are not. There are 53 samples of this kind, so the recognition rate of gear fault type is 97.5%.

In conclusion, compared with BoW model, the AEBoW model shows great advantages in the fault diagnosis under multi-operating conditions. Especially in the recognition of rotating speed, the performance of AEBoW model is much better.

(B) Operating efficiency

The operating efficiency of a diagnosis method directly affects its practical value. In order to illustrate the efficiency of the proposed method, the operating time of all test samples is tested. The computing platform is Ubuntu 16.04 + Opencv, the programming language is C++, and the CPU is Intel (R) Core (TM) i5-7200U.

The diagnostic process can be divided into three steps. The average time of each step of AEBoW model and traditional BoW model for one sample is calculated as shown in Table 3. Among them, the signal conversion of the two models is the same. In addition, three features are extracted simultaneously in the AEBoW model, so only the total time is calculated.

As shown in Table 3, for the fault diagnosis of a vibration signal, the average time of AEBoW model is 0.307 s, while BoW model is 0.222 s. Each sample is collected in 1 s. Therefore, both AEBoW and BoW are very efficient and have the ability to meet the real-time requirements.

In detail, the fault diagnosis of the third layer of AEBoW model takes the longest time. This process includes calculating Unoriented-SIFT word frequency vector and the distances between it and the vectors in Unoriented-SIFT codebook. Therefore, fault

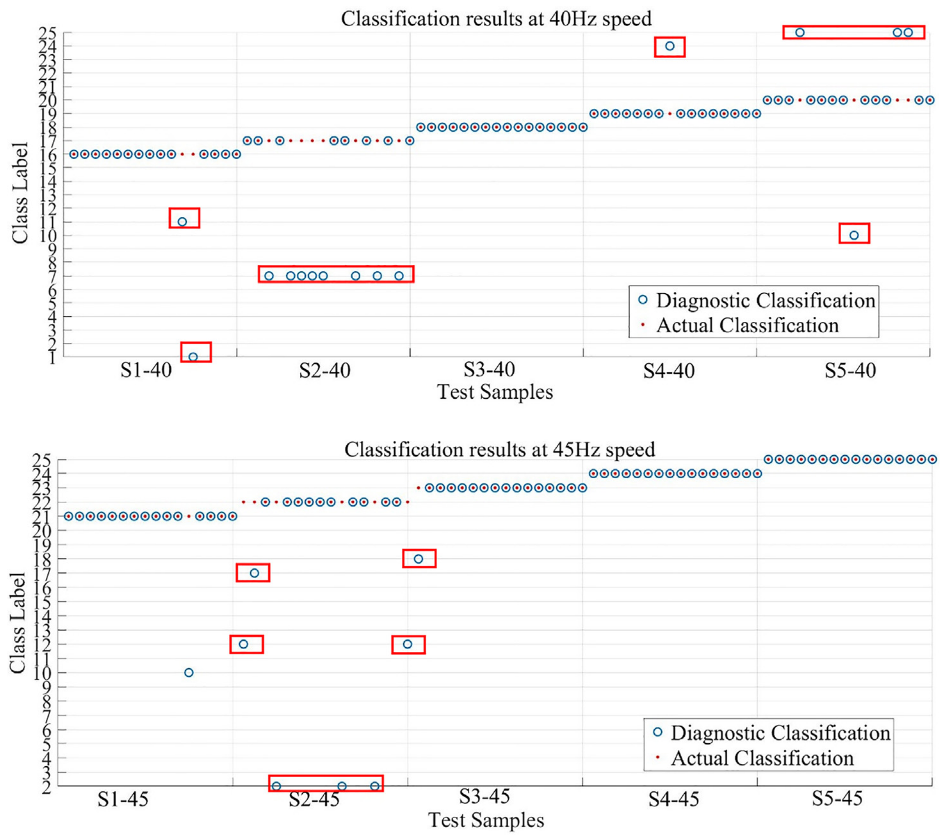

Fig. 11 (*continued*)

Table 3
The time-use comparison between AEBoW model and BoW model.

	Signal conversion /s	Feature extraction/s			Fault diagnosis/s		
AEBoW	0.069	LK	L2DIE	US	Layer 1	Layer 2	Layer 3
		0.016			0.050	0.053	0.119
BoW		0.015			0.138		

diagnosis should be completed in the first two layers as far as possible. According to Table 2, 5.20% and 8.12% of samples are diagnosed in the first two layers, which takes 0.135 s and 0.188 s for one sample, respectively. Therefore, the calculating efficiency of these samples is higher than that of BoW model.

5.2. Discussion on the selection of gray level

Gray level determines the value range of each pixel, so a larger value range means a higher representation accuracy for vibration amplitudes. However, according to Eq. (5), feature point is the pixel whose gray-scale value differs from the surrounding pixels. Therefore, the gray level needs to be enough to distinguish the vibration amplitudes, but it is not necessary to choose a very large gray level.

For the convenience of programming, OpenCV is used to extract features. In OpenCV, the data type of gray image could be 8 bit and 16 bit, because only these two types are unsigned. Half the space of 32 bit is negative, so 32 bit is equal to unsigned 16 bit. Therefore, the gray level could be 256 (8 bit) and 65,536 (16 bit). Other gray levels need to be obtained by degrading these two gray levels, which is meaningless. All test samples are converted into 16 bit images and then brought into the AEBoW model. The comparison results with 8 bit images are shown in the Tables 4–6.

Table 4
Time use.

	Signal conversion/s	Feature extraction/s
8bit	0.069	0.016
16bit	0.066	0.017

Table 5
Average number of extracted features.

	Broken	Normal	Root	Short	Wear
8bit	125.0	174.3	94.1	103.7	137.9
16bit	126.3	175.5	95.1	104.7	139.4

Table 6
Recognition rate of 7-fold cross-validation experiment.

	1-fold	2-fold	3-fold	4-fold	5-fold	6-fold	7-fold	Average
8bit	99.60%	99.60%	98.40%	98.80%	100%	98.80%	99.20%	99.2%
16bit	98.80%	99.60%	98.80%	98.80%	98.40%	98.40%	99.20%	98.85%

Table 4 shows that the time used for signal conversion and feature extraction of 16 bit is about equivalent to that of 8 bit images. The reason is that the larger value range does not increase the complexity of the algorithm. Table 5 shows the average number of extracted features of a sample for different fault types. The average number of extracted features of 16 bit is a little more than that of 8 bit. However, in the 7-fold cross-validation experiment, the average recognition rate of 16 bit is a little less than that of 8 bit.

In terms of operation efficiency, feature extraction and recognition rate, 16 bit has no obvious advantages, so 256 Gy level is enough for the working conditions described in this paper. It is suggested that researchers choose the appropriate gray level according to the complexity and accuracy of their data.

6. Conclusion

In actual work, the operating conditions of planetary gear are usually in a state of change, such as changes of rotating speed, but the traditional BoW model is not suitable for fault diagnosis under multi-operating conditions, because its recognition efficiency and accuracy decrease with the increase of fault types. Besides, the number of basic words in BoW needs to be set manually, which makes it difficult to get the optimal number. Therefore, a fault diagnosis method for planetary gear under multi-operating conditions based on adaptive extended bag-of-words (AEBoW) model is proposed in this paper. The sum of squared errors (SSE) is defined to describe the intra-class dispersion, and the optimal number of basic words is obtained by the gradient descent of SSE. The AEBoW model has three layers, and codebooks of each layer are constructed with different features. The probabilities that a signal belongs to each fault type are calculated in each layer, and the final diagnostic result is obtained by screening the classification results with higher probability layer-by-layer.

Traditional feature extraction methods are to highlight the fault features through noise reduction, but it is very difficult to extract weak fault features from planetary gear vibration signals by denoising. In this paper, a feature extraction method without denoising is proposed. After converting the raw vibration signals into gray images, the feature points are extracted directly from images, and the local kurtosis (LK), local 2-dimensional information entropy (L2DIE) and Unoriented-SIFT descriptor of each feature point are calculated. LK and L2DIE are proposed in this paper, which reflect the impact features and the complexity of vibration signals respectively. The three features are used to construct codebooks of each layer in AEBoW model. Compared with BoW model, the advantage of AEBoW model in fault diagnosis of planetary gear under multi-operating conditions is verified. In the experiment, the diagnostic accuracy of 25 fault states (5 rotating speeds and 5 gear fault types) reaches 99.5%. Without considering whether the rotating speed is accurately classified, the accurate recognition rate of the gear fault types reaches 100%. In addition, the experiment proves that this method is efficient enough to meet the real-time requirement, which has good application prospects.

In the future, several issues need to be further studied. First, a more efficient method of selecting the optimal number of basic words needs to be found. Second, this paper aims at the problem of machine switching in multiple working conditions, but the working conditions of some machines are changing continuously. Therefore, it is necessary to study how to improve this method to adapt to the fault diagnosis under continuous changing of working conditions. Last, it is worthwhile to study how to update codebooks in the BoW model automatically with self-learning technology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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